DALAS Project
Drug Repurposing

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1 Introduction

1.1 Problem Statement

Clearly define the problem your project addresses. Explain its relevance in the scientific or practical context.

The discovery of new drugs is, although crucial, an extremely expensive and time-consuming process. Many candidate drugs ultimately turn out to be not only ineffective, but also potentially toxic. It is thus of interest to repurpose already approved drugs for diseases they have not yet been tested against. To prioritize potential clinical trials, in addition to experts' knowledge, it's important to be able to predict the likelihood of success for such studies. This project aims to explore the data and develop a potential prediction pipeline for this purpose. We acknowledge that industrial and academic drug development employ far more comprehensive methods, this project, by contrast, focuses on a preliminary, data-driven approach and is exploratory in nature.

1.2 Motivation and Objectives

Explain why this problem is important. State your main objectives and hypotheses.

The ability to predict the success of drug repurposing has academic and clinical value, it helps reducing the experimental cost and accelerating the discovery of effective treatments. The main objectives of this project are to:

- Explore public datasets to identify some relevant features
- Develop a preliminary machine learning pipeline for success prediction
- Evaluate the model performance
- Identify biases and potential ways for improvement

We decided to specify a group of closely related diseases - immune system disorders and drugs that are approved for at least one member of this group.

2 Data

2.1 Data Sources

Describe where the data comes from, collection methods, and aggregation process.

2.2 Data Quality and Preprocessing

Discuss missing values, outliers, normalization, feature encoding, or transformations.

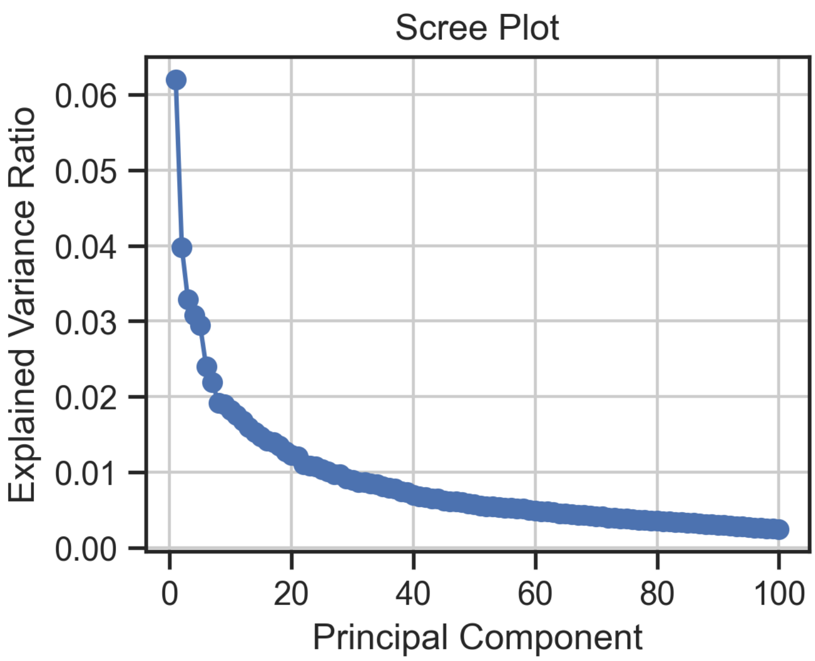

Figure 1: Scree plot of PCA on merged data.

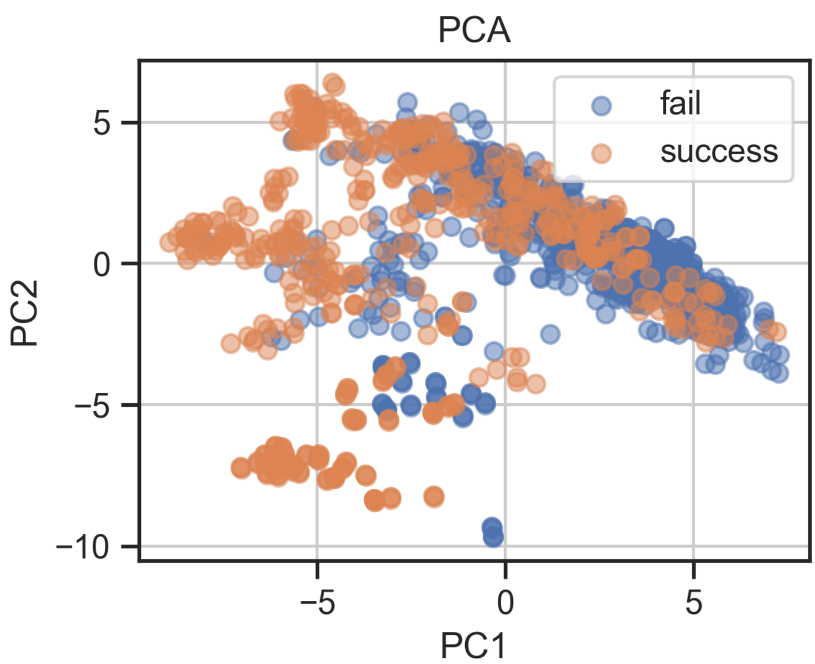

Figure 2: Scree plot of PCA on merged data.

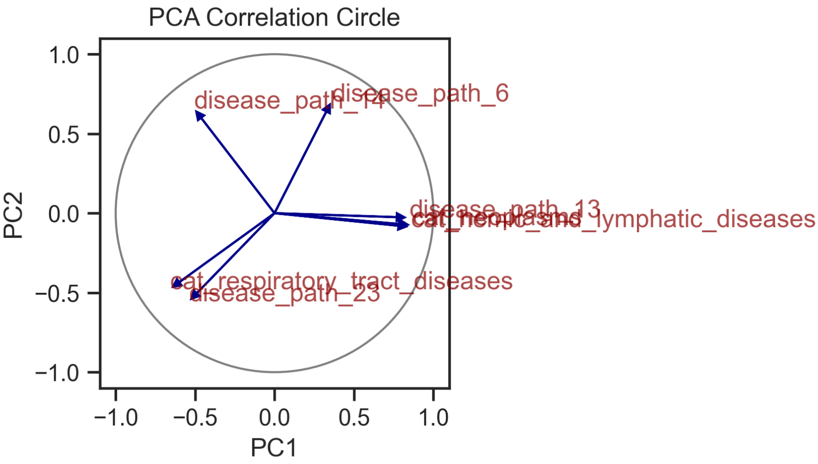

Figure 3: Scree plot of PCA on merged data.

2.3 Feature Description

Provide a table or list explaining each feature and its meaning. Optionally, include relationships between features.

```
    HeteroData(
  drug={ x=[742, 127] },
  disease={ x=[167, 25] },
  pathway={ x=[1509, 1] },
  (drug, interacts, pathway)={ edge_index=[2, 31266] },
  (disease, associated_with, pathway)={ edge_index=[2, 13608] },
  (drug, tryed_against, disease)={
    edge_index=[2, 1298],
    edge_label=[1298],
  }
)
```

3 Methods

3.1 Model Overview

Briefly describe the models used. Include general principles (e.g., linear models, random forests, neural networks).

3.2 Implementation Details

Mention important parameters, software/libraries used, and cross-validation setup.

4 Results

4.1 Model Comparison

Compare models quantitatively (accuracy, RMSE, AUC, etc.) and qualitatively (strengths/weaknesses).

4.2 Feature Analysis

Analyze the impact of key features on model performance (e.g., feature importance, correlation).

4.3 Failure and Success Cases

Highlight scenarios where the models perform well or poorly. Include illustrative figures or tables.

5 Discussion

Interpret the results. Relate findings to your original objectives. Discuss limitations and possible improvements.

6 Conclusion

Summarize your work, key findings, and potential future directions.

References